

How to measure the economic health cost of wildfires – A systematic review of the literature for northern America

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Abstract. There has been an increasing interest in the economic health cost from smoke exposure from wildfires in the past 20 years, particularly in the north-western USA that is reflected in an emergent literature. In this review, we provide an overview and discussion of studies since 2006 on the health impacts of wildfire smoke and of approaches for the estimation of the associated economic cost. We focus on the choice of key variables such as cost estimators for determining the economic impact of mortality and morbidity effects. In addition, we provide an in-depth discussion and guidance on the functioning, advantages and challenges of *BenMAP-CE*, freely available software of the US Environmental Protection Agency (EPA) that has been used in a growing number of studies to assess cost from wildfire smoke. We highlight what generates differences in outcomes between relevant studies and make suggestions for increasing the comparability between studies. All studies, however, demonstrate highly significant health cost from smoke exposure, in the millions or billions of US dollars, often driven by increases in mortality. The results indicate the need to take health cost into account for a comprehensive analysis of wildfire impacts.

Additional keywords: BenMAP-CE

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Introduction

Total area burned through wildfires in the United States has increased since the mid-1980s, with the north-west USA as one of the most affected regions (Spracklen *et al.* 2009; Liu *et al.* 2016; McClure and Jaffe 2018). Although there is some evidence to suggest this increase is due to a more normal fire mitigation regime in areas where fires had previously been suppressed (Doerr and Santin 2016), there exists a growing consensus that climate change has led and will lead to an increased incidence of wildfires (Fried *et al.* 2008; Spracklen *et al.* 2009; Flannigan *et al.* 2013; Liu *et al.* 2016). Abatzoglou and Williams (2016) found anthropogenic climate change to have contributed to increased fuel aridity and a doubling in the cumulative forest fire area since 1984, the main drivers of this being earlier spring snow melt and increasing spring and summer temperatures (Spracklen *et al.* 2009). Another driver of wildfires is the increased wildland-urban interface, the zone between unoccupied land and human population, where homes are built near or among lands prone to wildfires (Radeloff *et al.* 2018). Liu *et al.* (2016) estimate ‘more than 82 million individuals will experience on average a 31% to 57% increase in frequency and intensity’ of wildfire in the USA. This increase is to be greatest in the Pacific Northwest, with a 78% increase in annual mean area burned by 2050 (Spracklen *et al.* 2009). Linked to this, Cascio (2018) found that the size of the population at risk from smoke is going up: an increase in wildfire

prevalence has led to an increase in exposure to harmful pollutants emitted by wildfires like particulate matter (PM), and ground-level ozone (O₃). As such, interest in the cost to health of these wildfires is growing.

At present, there exists significant literature surrounding wildfires’ ecological impact (Reid *et al.* 2016); yet literature surrounding the valuation of indirect health costs associated with wildfires is still relatively new. Direct health costs from wildfire relate to damage such as death resulting from burn injuries whereas the indirect costs are the result of secondary impacts. Many general cost studies on wildfires vastly underestimate the true cost of those fires as they do not incorporate smoke-induced health costs. The literature suggests that these indirect costs may be much greater than assumed in the past and growing. It is estimated that between 2010 and 2015, wildfire smoke costs grew on average by 217% each year in the western USA (Jones and Berrens 2017).

Kochi *et al.* (2010) carried out a literature review on the economic costs of health impacts in 2010. At that time, they identified a need to better understand both the major and minor impacts of wildfire smoke on adverse health outcomes including more knowledge on aversion behaviours. In the 10 years since then, several new studies on the cost of health impacts from wildfires have emerged, particularly in the USA. This could in part be due to the increasing severity of wildfires, especially in the USA, but also due to the availability of *BenMAP-Community*

Edition (BenMAP-CE) or the ‘Environmental Benefits Mapping and Analysis Program’: open-source software provided by the US Environmental Protection Agency (EPA) that makes the evaluation of changes in air quality and the application of cost–benefit analysis in the wildfire context more accessible and has been used by several studies.

In the present literature review, we will examine the latest studies on the economic costs of adverse health-related impacts from wildfires. Although there is some literature on the health effects from prescribed fire smoke (Nguyen *et al.* 2020), we are not aware of economic valuation studies for prescribed fires.

We focus mostly on northern America, as the bulk of recent literature focuses on this geographic region but also include studies from other regions where relevant. We believe this to be a valuable exercise to inform future research and policy. Often-times, health impacts from smoke are still not considered when estimating the total cost from a wildfire as it is time-consuming. Synthesising the literature related to health costs allows us to better contextualise other costs from wildfires and to inform discussions on what drives costs from wildfires. As such, a better understanding of total wildfire costs can provide an indication where to focus mitigation spending for wildfire impacts.

We provide readers with practical guidance on how *BenMAP-CE* functions and where its challenges lie that makes the standardisation across *BenMAP-CE* studies difficult. We also discuss options on how to make those studies potentially more uniform for future research. The first part of the paper includes all studies that have been identified as relevant to the topic, whereas the second part focuses on studies that used *BenMAP-CE*.

This remainder of this paper is structured as follows: *The pollutants from wildfires that affect human health* section reviews the literature in the field, including the epidemiological effects of wildfire smoke, describing first the pollutants, and then health impacts of these pollutants. The *Economic cost of health impacts* and *BenMAP-CE* sections discuss literature that values health impact incidences to derive economic cost with a focus on *BenMAP-CE*. The final two sections provide a discussion and conclusion respectively.

Methodology

The literature search was conducted using search engines (Google Scholar and ResearchGate) and databases (CSIRO Publishing and Elsevier) in order to identify peer-reviewed articles as well as grey literature that tackle the subject. Through the citations and references of these papers, further evidence was found to contribute to this literature review. Search terms centred around wildfire/forest fire, prescribed fires/controlled burns, particulate matter (PM_{2.5} and PM₁₀), ground-level ozone (O₃) and health impacts/effects. Through the search, a focus was placed on literature from 2006 onwards as well as a concentration on North America. However, literature that added value was also utilised as supporting evidence even if not under these criteria. A filtering process – through titles, then abstracts and finally full papers – was then used to identify the relevant studies. We identified 47 studies as relevant, of which 12 estimated the economic cost of health impacts and of those, five used *BenMAP-CE*.

The pollutants from wildfires that affect human health

Wildfires generate smoke plumes that are the primary cause of indirect health costs. Smoke plumes are defined as ‘gases, smoke and debris that rise slowly from the fire while being carried along the ground because the buoyant forces are exceeded by those of the ambient surface wind’ (National Wildfire Coordinating Group 2018). These plumes dramatically alter air quality in the affected area and can increase both the levels of ozone (O₃) and PM to dangerous levels (Larsen *et al.* 2018). Other combustion products produced by wildfires include: polycyclic aromatic hydrocarbons (PAHs), carbon monoxide (CO), nitrogen oxides (NO_x), volatile organic compounds (VOCs) and sulfur dioxide (SO₂) (Hänninen *et al.* 2009).

There is consensus that wildfire smoke has a significant impact on both acute and long-term health endpoints in the exposed populations (US Forest Service 2018). In general, the health literature shows associations between smoke exposure and an increased risk of mortality (death) (Morgan *et al.* 2010; Johnston *et al.* 2012; Kochi *et al.* 2012; Shaposhnikov *et al.* 2014; Reid *et al.* 2016). Specifically, Johnston *et al.* (2012) found the estimated global mortality attributable to wildfire to be an average of 339 000 deaths annually across the world (the most affected regions being sub-Saharan Africa and south-east Asia). In addition, there is also evidence of the increased risk of morbidity (Kochi *et al.* 2010).

The evidence on health impacts from prescribed fires is very limited but literature on PM_{2.5} concentrations suggests they may be less severe than from wildfires. Generally, even though the burnt areas from wildfires and prescribed fires are similar in the USA per year, the measured PM_{2.5} concentrations are much lower for the latter (Jaffe *et al.* 2020). This is due to fuel types and fire management practice that avoids burning canopy or duff fuels in prescribed fires.

Particulate matter (PM)

Particulate matter is the catch-all term for ‘solid particles and liquid droplets found in the air’ and includes both PM₁₀ (diameters 10 µm and smaller) and PM_{2.5} (2.5 µm and smaller) (EPA 2018a). Fine PM consists of soot, organic carbon compounds and inorganic ash (Hänninen *et al.* 2009). Owing to its size, PM infiltrates deep into the lungs, with some even entering the bloodstream. Decreases in the levels of PM_{2.5} have been seen nationwide in ambient air as anthropogenic emissions (emitted by combustion engines) have been reduced. Fann *et al.* (2018a) found the estimated fraction of deaths in the USA due to PM_{2.5} from all sources to be 6.1% in 2005 (150 000 deaths), which fell to 4.6% (121 000 deaths) by 2014 as PM_{2.5} concentrations declined. Yet areas with a high wildfire prevalence have seen increases in levels of PM_{2.5} (particularly the western and north-western USA) attributable to the impact of wildfire (Liu *et al.* 2016; McClure and Jaffe 2018). In addition, Wegesser *et al.* (2009) demonstrated that PM_{2.5} produced by wildfire is of a greater toxicity than that found in ambient air, highlighting the need for health analysis of PM_{2.5} caused by wildfires.

The effects of PM_{2.5} are both acute and long term, with Adetona *et al.* (2016) describing PM_{2.5} as the most harmful pollutant from wildfire to public health.

Multiple reviews of the literature find a strong association between exposure to wildland fire smoke and respiratory morbidity and mortality (Youssef *et al.* 2014; Liu *et al.* 2015; Reid *et al.* 2016; Cascio 2018). This includes aggravated asthma, chronic obstructive pulmonary disease (COPD), bronchitis, decreased lung functioning and pneumonia. Regarding the exacerbations of asthma, studies including Henderson *et al.* (2011) and Morgan *et al.* (2010) found an association with increased respiratory mortality in a study of bushfire smoke in Australia (concentrating on the urban area of Sydney), a finding supported by a longer-term Australian study (Johnston *et al.* 2011). Meanwhile, an analysis of cities across Europe also found increases in respiratory mortality to be associated with increases in levels of PM₁₀ (Faustini *et al.* 2015).

Wildfire smoke exposure has also been shown to lead to higher rates of respiratory hospital admissions as a result of smoke exposure (Delfino *et al.* 2009; Lee *et al.* 2009; Morgan *et al.* 2010; Henderson *et al.* 2011; Rappold *et al.* 2011; Martin *et al.* 2013) and an increase in the rates of physician visits (Yao *et al.* 2016).

The research on PM_{2.5} exposure and cardiovascular morbidity shows mixed results. Haikerwal *et al.* (2015) is one of few studies to show an association between PM_{2.5} and cardiovascular outcomes. In that study, PM_{2.5} exposure was linked to an increased risk of out-of-hospital cardiac arrests and ischaemic heart disease (IHD) during the 2006–07 wildfires in Victoria, Australia. Reid *et al.* (2016) conclude that ‘too many studies and too many inconsistencies in findings exist to determine whether wildfire smoke exposure is associated with specific cardiovascular outcomes’.

Tropospheric ozone (O₃)

Another key pollutant from wildfires is tropospheric ozone (O₃). O₃ – a secondary pollutant – is generated by the photoreaction of NO_x (oxides of nitrogen) and VOCs, which are both produced by wildfires.

Studies by Jaffe and Widger (2012) and McClure and Jaffe (2018) have shown wildfire to contribute to an increase in the level above the ozone air quality standard. Unhealthy levels of ozone were shown to be 3.3 times more likely to occur on ‘smoke plume days’ in a study of a 2006–13 wildfire by Larsen *et al.* (2018), with the most heavily impacted areas being urban owing to the increased presence of non-methane organic compounds (NMOCs), which need to be present to produce O₃. Having said that, measuring O₃ production from wildfires is quite complicated as it is a secondary pollutant and as such, whereas particulate matter concentration tends to decrease away from the site of the fire, O₃ mixing ratios have the potential to increase (Jaffe and Widger 2012). Studies showing increased O₃ concentrations due to wildfires include Bossioli *et al.* (2012) and Kang *et al.* (2014).

Likely owing to the difficult measurement of O₃ due to wildfires, information on its impact on health is much sparser relative to PM (Reid *et al.* 2016). Reisen *et al.* (2015) state that although O₃ concentrations close to fires are unlikely to be high enough to cause concern, higher O₃ concentrations can occur ‘where concentrations are already high owing to anthropogenic activities’, for example in urban areas.

If concentrations are sufficiently elevated, ozone can have serious health implications, particularly for vulnerable populations and it is also a greenhouse gas. O₃ has been shown to irritate lungs and aggravate bronchitis and asthma, as well as leading to short-term mortality (occurring within less than 90 days) (Youssef *et al.* 2014; Reisen *et al.* 2015).

In addition, VOCs themselves (which react with nitrogen oxide to form ozone) have been found to be associated with symptoms like skin and eye irritation, drowsiness, coughing and wheezing (Youssef *et al.* 2014).

Population segmentation

The population segments most affected by wildfire smoke are the elderly, smokers and those with pre-existing conditions (Youssef *et al.* 2014). Also, increases in exposure and health risks are greater for women than men, and greater for African Americans, and in counties with [relatively] lower levels of education (Liu *et al.* 2017). These studies raise significant questions about inequalities leading to vulnerabilities of populations particularly with regard to ability or knowledge to implement preventative measures.

Another health concern associated with wildfires is birth outcomes. One study found the 2003 southern Californian wildfires to be associated with a reduced average birthweight (Holstius *et al.* 2012). These findings were confirmed by Cândido da Silva *et al.* (2014) and Flannigan *et al.* (2013).

Economic cost of health impacts

Health impacts from wildfires carry costs from lives lost, medical treatment, working time lost and disutility – for example from pain. Our literature search brought up only a small number of directly relevant papers estimating this economic cost. For much of the literature, emphasis is placed on western USA wildfires (e.g. Douglass 2008; Kochi *et al.* 2012; Richardson *et al.* 2012; Jones *et al.* 2016; Jones and Berrens 2017). Many of the studies use the cost–benefit transfer software tool *BenMAP-CE* in some form or another (Douglass 2008; Jones *et al.* 2016; Jones and Berrens 2017; Fann *et al.* 2018b). To measure mortality (lives lost) cost from wildfires, studies used the Value of Statistical Life (VSL). To capture morbidity cost (disease), either Cost of Illness (COI) or Willingness to Pay (WTP) or both were applied. The concepts of VSL, WTP and COI in the context of wildfires are discussed in the next section.

Measuring the cost of premature mortality – the value of a statistical life (VSL)

Costs from mortality are far greater than from any other health endpoint owing to the large cost associated with one death measured by the VSL, which is a measure of per-unit cost of premature mortality. The VSL refers to society’s aggregated WTP to save one anonymous person’s life and is derived from an individual’s WTP for a specific mortality risk reduction. For example, individuals may be asked how much they are willing to pay to reduce their risk of dying of 1 in 10 000 (or 0.0001) from a specific pollutant. This implies that for every 10 000 individuals, we would expect one individual to die if no reduction in the pollution occurs. If the average WTP for this 1/10 000 mortality risk reduction is US\$100 (all dollar values are in 2019 dollars

unless stated otherwise to allow a better comparison), then the value of VSL can be calculated as individual's average WTP for the risk reduction of x over x ($\text{US\$}100/0.0001 = \text{US\$}1 \text{ million}$) (EPA 2018b). Kochi *et al.* (2010) states that VSL estimates in the literature are between US\$2.3 and 16.4 million (2019 dollars); the EPA (2019a) provides a central estimate of US\$9.38 million (2019 dollars), which is also used in *BenMAP-CE* and derived from 26 VSL studies. The discussion about appropriate VSL is ongoing. For example, Rittmaster *et al.* (2006) lowered the VSL estimate in a study on wildfire smoke in order to reflect the group to be most likely affected by air quality changes, namely the elderly. Elderly people have fewer remaining years of life than other age groups; they may therefore be willing to pay less to reduce the mortality risk. One could argue, however, they may be willing to pay more because the remaining years of life are a scarce good to them (Kochi *et al.* 2012).

All studies that estimate the cost from mortality rely on VSL and on values provided by the EPA (or the Canadian equivalent) for different regulatory analyses. Generally, costs from even a small increase in premature mortality are far greater than any other cost associated with wildfire health impacts, with the next largest impact being from working days lost (Rittmaster *et al.* 2006; Kochi *et al.* 2010; Jones and Berrens 2017). However, the wide range of VSL estimates also makes it harder to contextualise the magnitudes of mortality cost across studies.

Rittmaster *et al.* (2006) found a 1-day increase in PM during a 2001 fire in Chisholm, Alberta, Canada, had aggregated health impacts of CAN\$18.5 million, most of which was from premature mortality. Similarly, Kochi *et al.* (2012) estimated the value of an additional 133 deaths due to cardiorespiratory illness from smoke exposure during 2003 southern Californian wildfires to be between US\$207.28 million and US\$2.08 billion. In a rare US east coast study, Rappold *et al.* (2014) examined a wildfire smoke episode in rural North Carolina. The purpose of this study was to look at the impact of potential interventions (like adequate forecasting), and they found the economic benefits of effective interventions to be US\$58 million in total, again mainly owing to avoided premature mortality. Jones and Berrens (2017) investigated wildfires in the entire western US from 2005 to 2015 and found that although emergency room visits were dominating in terms of numbers, mortality accumulated the greatest health cost at US\$1.9 billion (averaging \$177 million year⁻¹). In another similar longer-term study, Fann *et al.* (2018b) looked at the impact of wildfire episodes in the USA from 2008 to 2012. They found a significant increase in hospital admissions and premature deaths valued at between US\$13 and 24 billion per year for short-term exposure (within 5 years) and US\$90 and 154 billion per year for long-term exposure (over 5 years).

Measuring the cost from morbidity – deriving Willingness to Pay and Cost of Illness

Various approaches have been used to estimate the cost of morbidity impacts, which is challenging as the severity and duration of adverse health outcomes vary significantly (Kochi *et al.* 2010). Table 1 provides a summary of the COI and WTP estimates and their sources in the reviewed studies.

The COI approach only looks at direct costs from morbidity: it sums the resource and opportunity costs from being sick – the

treatment cost and lost wages – without considering the disutility associated with pain, discomfort and lower quality of life (also known as reduced QALY). The cost of the preventative action taken to avoid becoming sick is also not taken into account (e.g. the cost of buying an air cleaner). Generally, this approach is used to value the cost of specific health outcomes involving medical care or some form of direct expense.

Therefore, most COI estimates are derived from actual treatment data for specific endpoints from hospital and emergency departments, as shown in Table 2. Those figures range from less than US\$100 to close to US\$70 000 depending on the endpoint. Some studies include opportunity cost (wages lost) as part of COI. Work days lost are a major factor in morbidity costs, accounting for 36–74% of health costs (Kochi *et al.* 2010). In addition, there are COI that represent the average cost of illness per exposed person per day that are non-endpoint specific, e.g. US\$11.43 in Richardson *et al.* (2012), US\$3.60 in Richardson *et al.* (2013). The latter study also includes a COI estimate that considers lost leisure time, arriving at a COI of US\$20.10. Most of the *BenMAP-CE* studies apply the software's in-built COI values (Rittmaster *et al.* 2006; Rappold *et al.* 2014; Jones *et al.* 2016; Parthum *et al.* 2017). The COI estimates that are per exposed person (often obtained by survey data) would include individuals who suffered from serious cardiovascular conditions (caused by smoke) and those who purchased a face mask for protection. The latter would not show up in the hospital or Emergency Department (ED) data, which makes it difficult to compare these COI estimates with endpoint-specific COI.

In contrast to COI, the WTP of an individual to avoid any negative impact from the fire includes all potential cost. This includes disutility from symptoms and cost of preventative action in addition to, as measured by COI, medical expenditures and lost wages (Richardson *et al.* 2012). WTP can be derived through revealed preferences (RP) or stated preferences (SP) methods. The former uses market transactions to estimate WTP, the latter usually surveys in which participants express their subjective WTP to prevent adverse endpoints.

The Defensive Behaviour Method (DBM) also known as the averting behaviour method (Grossman 1972) is an RP method, where the individual's averting and mitigating behaviours (e.g. using a home air cleaner, staying indoors) to reduce symptoms days are taken into account. In the DBM, a utility maximisation model is used to determine where the marginal benefits of reduced time spent sick equals the marginal cost of reduced sick time. The underlying idea is that the individual will not spend more on defensive action than the benefit they receive from it. The marginal cost of reduced sick time is determined by dividing the average cost of the defensive action (e.g. cost of an air cleaner) by the incremental effect of the defensive action (e.g. the effect of an air cleaner to reduce symptoms by 1 day), which is derived statistically.

Richardson *et al.* (2013) used DBM and the contingent valuation method (CVM) to determine the WTP of reducing symptom days by 1 day for the 2009 California Station fire. For CVM, an SP method, individuals affected by wildfire smoke are explicitly asked about their WTP to reduce the symptom days their household experienced by $x\%$. Jones *et al.* (2016)